

AI 620: Fundamentals of Data Engineering

Assignment 2

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Part 1: SVM Price Classification on PakWheels Used Car Data

Assignment Overview

This part of the assignment focuses on applying fundamental data engineering concepts to build a complete machine learning pipeline for binary classification. Using a real-world dataset of used car listings from PakWheels, you will implement various data engineering techniques including data cleaning, preprocessing, feature engineering, and pipeline construction.

Dataset Description

The dataset `data-a1.csv` contains used car listings with the following attributes:

Column	Type	Description
<code>addref</code>	<code>int</code>	Unique ad reference ID
<code>city</code>	<code>str</code>	City of listing
<code>assembly</code>	<code>str</code>	Local / Imported
<code>body</code>	<code>str</code>	Body type (Sedan, Hatchback, SUV, etc.)
<code>make</code>	<code>str</code>	Manufacturer (Toyota, Honda, Suzuki, etc.)
<code>model</code>	<code>str</code>	Car model name
<code>year</code>	<code>float</code>	Model year
<code>engine</code>	<code>float</code>	Engine capacity in cc
<code>transmission</code>	<code>str</code>	Manual / Automatic
<code>fuel</code>	<code>str</code>	Petrol / Diesel / Hybrid
<code>color</code>	<code>str</code>	Exterior colour
<code>registered</code>	<code>str</code>	Registration location
<code>mileage</code>	<code>int</code>	Odometer reading in km
<code>price</code>	<code>float</code>	Asking price in PKR (Continuous target)

Table 1: Dataset columns description

Learning Objectives

This part reinforces core data engineering concepts:

- **Data Quality Assessment:** Identifying missing values, outliers, and data issues
- **Data Cleaning:** Handling missing data, outliers, and inconsistent values
- **Feature Engineering:** Creating derived features from existing data

- **Pipeline Construction:** Building reusable preprocessing pipelines
- **Data Leakage Prevention:** Understanding and avoiding common pitfalls
- **Feature Selection:** Reducing dimensionality while preserving information
- **Model Evaluation:** Assessing model performance with appropriate metrics

Key Data Engineering Concepts Covered

1. Missing Data Handling

- Analysis of missing value patterns
- Imputation strategies (median for numerical, most frequent for categorical)
- When to drop vs. impute missing values

2. Data Transformation

- Creating binary target from continuous variable:

$$\text{HighPrice} = \begin{cases} 1 & \text{if price} \geq \text{median price} \\ 0 & \text{otherwise} \end{cases}$$

- Feature engineering:

$$\text{car_age} = \text{max_year_train} - \text{year}$$

$$\text{mileage_per_year} = \frac{\text{mileage}}{\text{car_age} + 1}$$

3. Pipeline Engineering

- Numerical pipeline: `SimpleImputer(median) → StandardScaler`
- Categorical pipeline: `SimpleImputer(most_frequent) → OneHotEncoder(handle_unknown=)`
- `ColumnTransformer` for mixed data types
- Chaining preprocessing with model training

4. Data Leakage Prevention

- Proper train/test split before any preprocessing
- Feature engineering using only training data statistics

Support Vector Machines (SVM) Overview

Support Vector Machines are powerful supervised learning models used for classification and regression tasks. In this assignment, you will implement two types of SVM classifiers:

Linear SVM

Linear SVM attempts to find a hyperplane that best separates classes in the original feature space. The optimal hyperplane maximizes the margin between the closest points of different classes (support vectors).

$$\text{Decision function: } f(x) = w^T x + b$$

Key characteristics:

- Works well when data is linearly separable or approximately linear
- Computationally efficient for high-dimensional spaces
- Provides a linear decision boundary
- Easier to interpret which features influence the decision

RBF (Radial Basis Function) SVM

RBF SVM uses kernel trick to transform data into a higher-dimensional space where it becomes linearly separable. The RBF kernel computes similarity between points based on their distance.

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

Key characteristics:

- Can capture complex, non-linear relationships in data
- More flexible decision boundary than linear SVM
- Requires tuning of hyperparameters (C and γ)
- Computationally more intensive than linear SVM
- Particularly effective when relationship between features and target is non-linear

SVM Requirements and Best Practices

- **Feature Scaling:** SVM is sensitive to feature scales. Features must be normalized/standardized before training (hence the `StandardScaler` in your pipeline)
- **Kernel Selection:** Linear kernel is faster and often sufficient for high-dimensional data; RBF kernel can capture complex patterns but requires more data and tuning
- **Probability Estimates:** Setting `probability=True` enables `predict_proba` method, useful for ROC curves and probability-based decisions

Part 1 Assignment Structure

1. Part 1-2: Data Loading and Initial Cleaning

- Load CSV and remove identifier columns
- Analyze high-cardinality and near-zero variance features
- Document data quality observations

2. Part 3-4: Data Inspection and Missing Values

- Generate summary statistics
- Visualize missing value patterns
- Justify imputation strategies

3. Part 5-6: Exploratory Data Analysis

- Analyze categorical variable distributions
- Examine numerical variable distributions
- Identify skewness and outliers

4. Part 7: Data Cleaning Pipeline

- Apply outlier removal thresholds
- Implement missing value handling
- Validate data quality improvements

5. Part 8-9: Target Creation and Data Splitting

- Create binary classification target

- Perform stratified train/test split
- Verify class balance

6. Part 10: Feature Engineering

- Create `car_age` and `mileage_per_year` features
- Apply transformations consistently
- Prevent data leakage

7. Part 11: Preprocessing Pipeline Design

- Design column-specific transformations
- Build reusable pipeline components
- Handle high-cardinality categoricals

8. Part 12-13: Model Engineering and Evaluation

- Train Linear and RBF SVM models
- Generate comprehensive evaluation metrics (classification reports, confusion matrices, ROC curves)
- Compare model performance

9. Part 14: Feature Selection Techniques

- Apply `VarianceThreshold` for low-variance filtering
- Implement Recursive Feature Elimination (RFE)
- Compare dimensionality reduction strategies

10. Part 15: Final Discussion

- Answer all the questions.

Note: The dataset `data-a1.csv` should be placed in the same folder as the notebook before execution.

Part 2: Web Scraping Fundamentals

Website: `http://books.toscrape.com/`

Total Points: 100

Assignment Overview

This part of the assignment focuses exclusively on web scraping using Python. You will build progressively sophisticated scrapers for `http://books.toscrape.com/`, learning essential techniques for data extraction from static websites.

Learning Objectives

- Send HTTP requests and parse HTML responses
- Extract structured data using BeautifulSoup
- Navigate pagination systems
- Handle errors and edge cases gracefully
- Implement ethical and polite scraping practices
- Save scraped data to CSV files

Part 2: Basic Scraping (30 points)

Task 1.1: Explore the Website (5 points)

Task Requirements:

- Send a GET request to `http://books.toscrape.com/`
- Check the status code of the response and print it
- Parse the HTML content using BeautifulSoup
- Extract and print the title of the page

Task 1.2: Extract Book Information from Homepage (15 points)

Task Requirements: Extract the following information for **all books** displayed on the first page:

- **Title** - from the `title` attribute of the `<a>` tag inside `<h3>`
- **Price** - extract text, remove the '£' symbol, and convert to float
- **Availability** - extract the text from the "In stock" element
- **Star rating** - convert text rating (One, Two, Three, Four, Five) to integers (1-5)
- **Relative URL** - extract the `href` attribute from the link inside `<h3>`

Store the extracted data in a pandas DataFrame and display the first 5 rows.

Task 1.3: Save to CSV (5 points)

Task Requirements: Save the DataFrame created in Task 1.2 to a CSV file named `books_homepage.csv`.

Task 1.4: Extract Book Details from Individual Pages (5 points)

Task Requirements: Choose any 3 books from your DataFrame. For each book:

1. Construct the full URL by combining the base URL with the relative URL
2. Visit the book's detail page by sending a new request
3. Extract the **product description** from the meta tag with `name="description"`
4. Extract the **UPC** from the product information table
5. Print the results for each book

Task 2: Pagination and Multiple Pages (30 points)

Task 2.1: Implement Pagination Logic (15 points)

Task Requirements: Write a function that scrapes all books from the first **3 pages** of the catalogue.

- Base URL pattern: `http://books.toscrape.com/catalogue/page-1.html`

- Pages increment by changing the number: page-1.html, page-2.html, page-3.html
- Extract the same fields as in Task 1.2
- Add a `page_number` column to track the source page
- Implement polite scraping: wait 1-2 seconds between page requests
- Return a DataFrame with all books from all pages

Task 2.2: Error Handling and Robustness (10 points)

Task Requirements: Modify your function to handle common errors robustly:

- Use try-except blocks for network errors and timeouts
- Check if the page exists (status code 200)
- Implement retry logic with exponential backoff
- Handle missing elements gracefully (use conditional statements)
- Print progress updates during scraping
- Continue scraping even if one page fails (track failed pages)

Create an enhanced version called `scrape_multiple_pages_robust()`.

Task 2.3: Save Combined Data (5 points)

Task Requirements: Save the scraped data from Task 2.2 to `books_multipage.csv` and verify the save was successful.

Task 3: Category-Based Scraping (25 points)

Task 3.1: Extract Category Links (10 points)

Task Requirements: Extract all book categories from the sidebar on the homepage:

1. Navigate to the homepage: <http://books.toscrape.com/>
2. Find the sidebar div with class `"side_categories"`
3. Extract all category names and their URLs

4. Exclude the "Books" category (which is just a header)
5. Store in a dictionary: `{category_name: category_url}`

Return a dictionary of all categories.

Task 3.2: Scrape a Complete Category (15 points)

Task Requirements: Choose **one** category (e.g., "Mystery", "Travel", or "Science Fiction") and scrape **all** books in that category, including all paginated pages.

Task 4: Data Aggregation and Summary (10 points)

Task Requirements: Load all three CSV files you've created and combine them into a single DataFrame:

- `books_homepage.csv`
- `books_multipage.csv`
- `books_category.csv`

Add a **source** column to identify where each record came from. Display the combined DataFrame info and shape.

Task 5: Reflection and Submission (5 points)

Overall Submission Guidelines

- Follow the naming convention: `<roll_number_AI620_ASSIGNMENT2_Part_X.ipynb>`
- Only LMS submissions will be accepted
- Complete all code blocks marked with "___"
- Answer all written response questions in markdown cells
- Run **Kernel** → **Restart & Run All** before submission
- Ensure all CSV files are generated and saved

Submission Notes:

Due: **March 11, 2026, 11:59 PM EST**

Total Points: **252** (Part 1: 152 points and Part 2: 100 points)

1. Your Jupyter Notebook file (.ipynb) containing both Part 1 and Part 2
2. All generated CSV files:
 - Part 1: Any output files from the SVM classification
 - Part 2: `books_homepage.csv`, `books_multipage.csv`, `books_category.csv`

Academic Integrity

- You are allowed to use AI tools to understand concepts or debug syntax errors
- You are **not allowed** to use AI to generate complete code or written responses
- You must clearly declare any use of AI at the end of your notebook
- Anyone found violating this policy will be awarded zero credit
- A viva may be conducted if misuse of AI is suspected

End of Assignment 2